

A Survey of Multimodal Retrieval-Augmented Generation for Chart and Document Understanding

InteractiveSurvey

Abstract

The exponential growth of multimodal documents has exposed critical limitations in traditional retrieval systems, which often fail to interpret the complex interplay between text, charts, and structured layouts. As large language models evolve, the primary bottleneck has shifted from generation capacity to the precision of retrieving grounded evidence from these heterogeneous sources. This survey comprehensively reviews Multimodal Retrieval-Augmented Generation frameworks specifically designed for chart and document understanding, bridging the gap between raw visual data and actionable knowledge. We systematically analyze advanced strategies that move beyond holistic image processing to dissect and index fine-grained visual elements within complex document ecosystems. Our findings highlight the superiority of late interaction mechanisms and generative embedding paradigms in preserving token-level details, alongside layout-aware graph constructions that enable sophisticated multi-hop reasoning across disjointed entities. Furthermore, we demonstrate how adaptive retrieval frameworks optimize computational resources by dynamically aligning query complexity with retrieval depth, ensuring accurate localization of evidence in high-density contexts. By synthesizing these disparate research threads, this work provides a unified roadmap for developing robust, visually grounded document intelligence systems capable of handling long-context, layout-rich data.

1 Introduction

The exponential growth of digital information has transformed documents from static text repositories into complex, multimodal entities integrating dense textual narratives, intricate data visualizations, and structured layouts. Traditional retrieval systems, predominantly designed for plain text, of-

ten struggle to interpret the nuanced interplay between charts, tables, and surrounding prose, leading to significant information loss during the extraction process. As large language models evolve to handle broader context windows, the bottleneck has shifted from generation capacity to the precision of retrieving relevant, grounded evidence from heterogeneous sources. Consequently, there is an urgent need for advanced frameworks capable of understanding and indexing the rich semantic and spatial relationships inherent in modern document ecosystems, ensuring that generated responses are both accurate and visually grounded.

This survey paper focuses on Multimodal Retrieval-Augmented Generation (RAG) specifically tailored for chart and document understanding, addressing the critical gap between raw visual data and actionable textual knowledge [1]. Unlike general multimodal RAG approaches that treat images as holistic units, this work examines specialized strategies designed to dissect and index fine-grained visual elements within complex layouts. We explore how recent advancements enable systems to bypass brittle optical character recognition pipelines, instead leveraging direct visual-textual alignment to capture subtle correlations in data visualizations and structured tables. By synthesizing current literature, this review elucidates the mechanisms that allow models to reason over long-context, layout-rich documents, providing a comprehensive roadmap for developing robust, next-generation document intelligence systems.

The core of this review begins by analyzing advanced multimodal retrieval and indexing strategies that redefine how visual and textual data are represented and aligned [2]. We examine the shift from early fusion methods to late interaction mechanisms, which preserve token-level and patch-level details to facilitate precise cross-modal matching without compressing inputs into lossy global vectors. Furthermore, we discuss the emer-

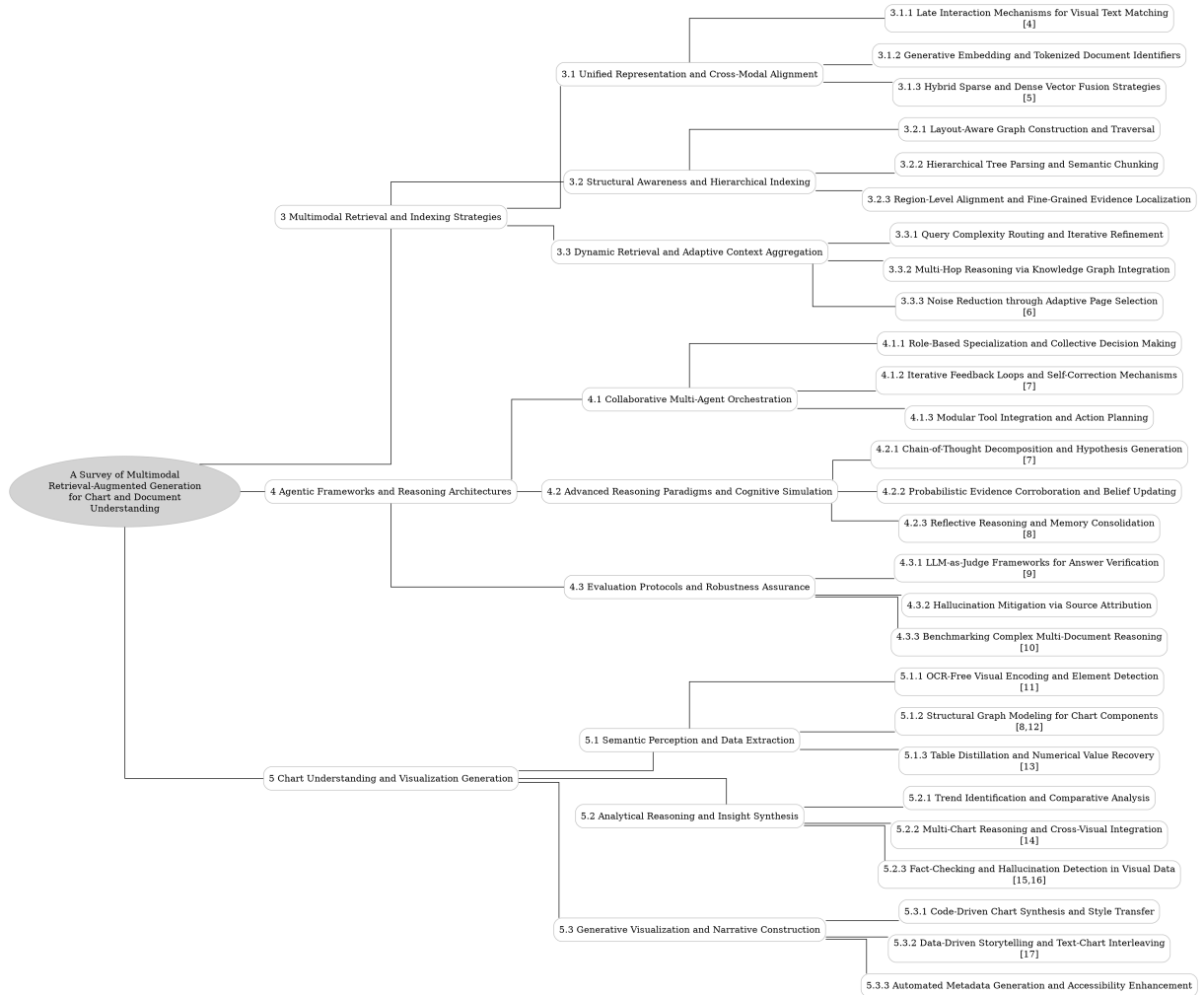


Figure 1: The outline of our survey: A Survey of Multimodal Retrieval-Augmented Generation for Chart and Document Understanding

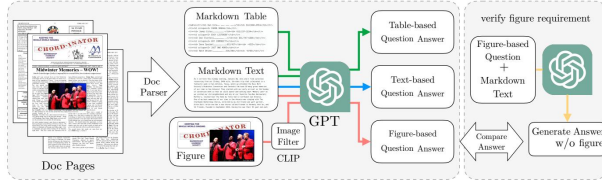


Figure 2: Chart from 'VisR-Bench: An Empirical Study on Visual Retrieval-Augmented Generation for Multilingual Long Document Understanding' (arXiv:2508.07493)

gence of generative embedding paradigms that utilize tokenized document identifiers to bridge the gap between user intent and document content, alongside hybrid fusion strategies that integrate sparse lexical precision with dense semantic understanding. These foundational techniques collectively enhance the system’s ability to retrieve relevant evidence from high-density visual contexts while maintaining computational scalability.

$$\mathcal{B}_{q_k}^- = \mathcal{B}^{\text{batch}} \setminus \mathcal{B}_{q_k}^+$$

Figure 3: Chart from 'LFRAG: Layout-oriented Fine-grained Retrieval-Augmented Generation on Multimodal Document Understanding' (arXiv:2605.22829)

Building upon these representation methods, we delve into structural awareness and hierarchical indexing techniques that capture the logical flow and spatial topology of documents. This section details how layout-aware graph construction and hierarchical tree parsing decompose pages into interconnected nodes, preserving long-range dependencies and nested relationships that flat chunking strategies often sever. By encoding cross-page associations and semantic hierarchies, these approaches enable sophisticated traversal mechanisms capable of multi-hop reasoning across disjointed visual and textual entities. We further explore region-level alignment protocols that ensure fine-grained evidence localization, allowing models to distinguish authentic modality-specific information from noise and accurately map atomic conclusions to their precise locations within complex layouts.

Finally, the survey investigates dynamic retrieval and adaptive context aggregation frameworks that optimize the interaction between query complexity and retrieval depth. We analyze sys-

tems employing query complexity routing to dynamically decompose high-level inquiries into multi-step plans, directing simple questions to lightweight indices while reserving graph-based traversal for intricate, multi-hop reasoning tasks. The discussion extends to iterative refinement processes where retrieval is treated as a recursive operation, utilizing feedback loops to expand search horizons and fill information gaps progressively. These adaptive mechanisms ensure that computational resources are allocated efficiently, enabling the synthesis of distributed information across diverse modalities to formulate comprehensive and contextually rich responses.

The primary contribution of this survey lies in its systematic categorization and critical analysis of state-of-the-art techniques specifically targeting the challenges of chart and document understanding within multimodal RAG frameworks [3]. By consolidating disparate research threads ranging from late interaction mechanisms to hierarchical graph traversal, we provide a unified perspective on how structural and visual nuances can be effectively leveraged for improved retrieval accuracy. This work not only identifies current limitations in handling long-context, layout-rich data but also outlines promising directions for future research, serving as an essential reference for researchers and practitioners aiming to build more robust, visually grounded document intelligence systems.

$$k = \left\lceil \frac{n}{b} \right\rceil$$

Figure 4: Chart from 'Vision-Guided Chunking Is All You Need: Enhancing RAG with Multimodal Document Understanding' (arXiv:2506.16035)

2 Multimodal Retrieval and Indexing Strategies

2.1 Unified Representation and Cross-Modal Alignment

2.1.1 Late Interaction Mechanisms for Visual Text Matching

Late interaction mechanisms have emerged as a pivotal architecture for visual text matching, addressing the limitations of early fusion by preserving fine-grained semantic details prior to aggregation. Unlike traditional approaches that compress visual and textual inputs into single dense vectors, late interaction maintains sequences of contextualized embeddings for both modalities. This design allows for token-level or patch-level alignment, enabling the model to compute similarity scores between every query token and every document visual patch. Such granularity is critical for document understanding tasks where precise localization of information within complex layouts, charts, or tables is required, effectively mitigating the information loss inherent in global pooling strategies.

Recent advancements adapt this paradigm to multimodal contexts by leveraging Vision-Language Models to generate multi-vector representations directly from document images [4]. Pioneering systems eliminate brittle Optical Character Recognition pipelines by treating image patches as distinct vectors, facilitating direct interaction between query text and visual regions. These models project visual tokens into a shared embedding space with textual queries, allowing for sophisticated cross-modal attention mechanisms. By deferring the interaction step until after independent encoding, these architectures achieve a balance between computational efficiency and retrieval accuracy, supporting scalable indexing of large document collections while retaining the capacity for nuanced, local feature matching.

The efficacy of late interaction lies in its ability to capture subtle correspondences that global representations often overlook, particularly in scenarios involving long-context documents or heterogeneous data types. Formal scoring functions aggregate maximum similarity values across token pairs, ensuring that relevant visual evidence is identified even when surrounded by noisy or irrelevant content. This approach significantly enhances performance in retrieval-augmented generation tasks, where the system must locate spe-

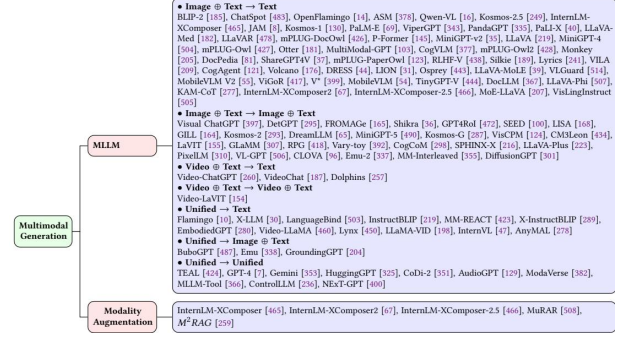


Figure 5: Chart from 'A Survey on Multimodal Retrieval-Augmented Generation' (arXiv:2504.08748)

cific visual-textual cues to formulate accurate responses. Consequently, late interaction has become a foundational component in state-of-the-art multimodal retrievers, driving improvements in both precision and robustness across diverse document domains.

2.1.2 Generative Embedding and Tokenized Document Identifiers

Generative embedding paradigms have evolved from static dense vectors to dynamic, tokenized representations that capture fine-grained semantic nuances. Unlike traditional bi-encoder architectures that compress entire documents into single fixed-dimensional vectors, generative approaches leverage decoder-only models to produce contextualized token sequences serving as document identifiers. This shift enables the model to implicitly learn complex structural dependencies and latent information needs by generating synthetic queries or identifiers that align closely with user intent, effectively bridging the gap between document content and retrieval objectives without relying on explicit parsing pipelines.

A prominent manifestation of this evolution is the adoption of late-interaction mechanisms within generative frameworks, exemplified by models like ColPali. These systems process document pages directly as images through vision-language models, extracting patch-level embeddings that function as soft tokens. Rather than collapsing these features into a global representation, the architecture preserves the multi-vector structure, allowing for token-level similarity computations during retrieval. This method eliminates the need for optical character recognition while maintaining high-fidelity interactions between query tokens and document visual patches, significantly enhancing performance on layout-rich and multi-

modal documents where spatial context is critical.

Furthermore, the integration of generative capabilities facilitates unified indexing strategies across heterogeneous modalities. By treating document chunks, tables, and images as sequences of tokens within a shared embedding space, these models support cross-modal matching and multi-hop reasoning over long contexts. The resulting tokenized identifiers enable scalable retrieval over millions of passages by optimizing the trade-off between computational efficiency and representational richness. Consequently, generative embedding not only improves retrieval accuracy through granular alignment but also simplifies the infrastructure required for managing diverse document types in large-scale knowledge bases.

2.1.3 Hybrid Sparse and Dense Vector Fusion Strategies

Hybrid sparse and dense vector fusion strategies have emerged as a critical paradigm for overcoming the limitations of unimodal retrieval in complex document understanding tasks. While dense embeddings excel at capturing semantic similarity through continuous vector spaces, they often struggle with exact keyword matching and rare entity resolution. Conversely, sparse methods like BM25 provide robust lexical precision but lack deep contextual understanding. Contemporary approaches address this dichotomy by integrating both representations, leveraging the complementary strengths of semantic density and lexical sparsity to enhance recall across heterogeneous data distributions.

The fusion mechanism typically operates through late-interaction frameworks where independent retrieval streams are aggregated using sophisticated ranking algorithms. A prevalent technique involves Reciprocal Rank Fusion (RRF), which normalizes and combines ranked lists from dense cosine similarity searches and sparse term-frequency indices without requiring extensive hyperparameter tuning for score calibration. More advanced architectures employ learnable gating mechanisms or hybrid scaling parameters to dynamically weight the contribution of each modality based on query characteristics. This adaptive weighting ensures that the system prioritizes lexical matches for factual queries while relying on dense vectors for conceptual reasoning, effectively balancing relevance and context.

Recent innovations extend these strategies into

the visual domain, adapting hybrid principles for multimodal retrieval augmented generation [5]. By constructing hybrid masks that preserve low-correlation detail features via dense masking while amplifying salient contexts through sparse masking, models can better handle high-density visual interactions and layout-aware chunking. These holistic retrievers fine-tune embedding spaces using composite loss functions that align sparse and dense objectives, significantly improving performance on benchmarks requiring precise cross-modal grounding. Consequently, hybrid fusion not only mitigates the parameter constraints of large language models but also establishes a robust foundation for scalable, accurate retrieval in multi-granularity document analysis.

2.2 Structural Awareness and Hierarchical Indexing

2.2.1 Layout-Aware Graph Construction and Traversal

Layout-aware graph construction initiates by decomposing document pages into heterogeneous nodes representing textual segments, tables, figures, and section headers, preserving their intrinsic spatial coordinates and semantic types. Unlike flat chunking strategies, this process employs structured-semantic agents to analyze multimodal cues, organizing elements into a human-intuitive hierarchical topology that mirrors the document's logical flow. By serializing complex structures like multi-column text and nested tables into arranged sequences, the system transforms unstructured inputs into a well-formed graph where nodes encapsulate both content and layout metadata, ensuring that critical visual-contextual relationships are not lost during ingestion.

The resulting graph structure explicitly encodes cross-page associations and long-range dependencies through directed edges that link related elements across document boundaries. This topology supports the modeling of nested sections, inter-referenced components, and procedural sequences, effectively creating a backbone for structural reasoning. As the graph is incrementally enriched page by page, it maintains a dynamic state that captures global document semantics alongside local chunk associations. This approach addresses the limitations of contextually incomplete retrievals by establishing explicit connections between disjointed visual and textual entities, thereby facil-

itating robust multi-hop reasoning over complex, lengthy documents.

Traversal mechanisms leverage this rich topology to execute sophisticated retrieval strategies that combine dense vector search with symbolic graph navigation. By decoupling vector indexing from graph traversal, systems can perform approximate nearest neighbor searches to identify candidate nodes before engaging in multi-hop exploration to synthesize information across modalities. This hybrid approach enables the resolution of complex queries requiring the integration of visual features from charts with textual evidence from distant sections. Consequently, layout-aware traversal significantly enhances performance on tasks demanding fine-grained spatial reasoning and the synthesis of distributed information, overcoming the perspective narrowing often observed in conventional retrieval-augmented generation pipelines.

2.2.2 Hierarchical Tree Parsing and Semantic Chunking

Hierarchical tree parsing addresses the critical limitation of traditional flattening methods, which fragment documents into isolated chunks and sever logical dependencies between paragraphs, tables, and visual elements. Instead of treating pages as independent units, this approach decomposes the document into a recursive structure where leaf nodes retain original layout attributes while intermediate nodes aggregate semantically related blocks across page boundaries. By utilizing clustering algorithms, such as Gaussian mixture models, text blocks are grouped based on semantic proximity, allowing the system to reconstruct the document’s inherent topology rather than relying solely on linear sequence or fixed token limits.

Semantic chunking within this hierarchy operates at multiple granularities to preserve context that is often lost in standard segmentation. At the fine level, chunks are defined by natural semantic boundaries rather than arbitrary character counts, ensuring that complex layouts and multi-column texts remain intact. These chunks are then recursively summarized to form higher-level nodes, creating a tree-organized index where the root node encapsulates the global topic semantics. This recursive abstractive processing enables the retention of both local details and global narrative flow, facilitating retrieval mechanisms that can navigate from high-level summaries down to specific evi-

dence without losing structural coherence.

The resulting tree structure supports advanced retrieval strategies by enabling query-aligned attention across different hierarchical levels. During inference, the system can dynamically emphasize either layout-level regions or fine-grained content based on the specific query intent, effectively bridging the gap between coarse document understanding and precise evidence extraction. This architecture significantly enhances reasoning capabilities for long-form documents, as it mitigates the context window constraints of large language models by providing structured, semantically dense pathways to relevant information. Consequently, the integration of hierarchical parsing with semantic chunking establishes a robust foundation for complex document question answering and multimodal reasoning tasks.

2.2.3 Region-Level Alignment and Fine-Grained Evidence Localization

Region-level alignment addresses the critical gap where conventional baselines underutilize visual modalities due to insufficient multimodal fusion mechanisms. Advanced frameworks now employ hierarchical parsing to preserve spatial and relational cues within charts and figures, ensuring that local coherence within sections aligns with the global document organization. By enforcing consistent heading hierarchies and leveraging layout-aware encoders, these methods capture discriminative visual-semantic signals that are often lost in naive, isolated extraction processes, thereby significantly improving accuracy on visually grounded reasoning tasks.

To achieve fine-grained evidence localization, recent approaches implement coarse-to-fine workflows involving specialized agents for hunting relevant regions and providing iterative feedback. This process often includes a two-stage filtering mechanism, starting with preliminary screening followed by rigorous cross-modal verification to validate that keypoints are retrievable from their source modality but not others. Such strategies effectively resolve evidence sparsity by distinguishing authentic modality-specific information from noise, ensuring that atomic conclusive statements are accurately mapped to their precise locations within complex document layouts.

Furthermore, addressing the challenge of cross-page reasoning requires integrating evidence scattered across multiple pages through multi-

granularity retrieval and graph-based navigation. Methods utilizing parent page retrieval and document-level summaries facilitate long-distance evidence integration, allowing models to connect logical relationships beyond surface-level semantic similarity. By constructing page graphs or employing tool-augmented "zoom-in" strategies, systems can dynamically aggregate fragmented evidence, enabling robust inference that spans heterogeneous modalities and diverse structural contexts within long-document understanding scenarios.

2.3 Dynamic Retrieval and Adaptive Context Aggregation

2.3.1 Query Complexity Routing and Iterative Refinement

Query complexity routing addresses the heterogeneity of user intents by dynamically decomposing high-level inquiries into multi-step, multi-granularity retrieval plans. Rather than relying on static tool invocation, advanced frameworks employ planner agents to analyze query difficulty and modality requirements, effectively routing simple questions to lightweight indices while directing complex, multi-hop queries toward sophisticated graph-based traversal mechanisms. This strategic orchestration mitigates the limitations of smaller models on intricate tasks by aligning retrieval depth with the specific reasoning demands of the input, ensuring that computational resources are allocated efficiently based on the structural complexity of the underlying document corpus.

Iterative refinement complements this routing by treating retrieval as a recursive process rather than a single-shot operation. Through mechanisms such as multi-hop graph expansion and query rewriting, systems progressively augment initial queries with semantically related neighbors discovered during traversal, thereby expanding the search horizon beyond immediate lexical matches. This approach often incorporates feedback loops where intermediate retrieval results are analyzed to identify information gaps, prompting the generation of refined sub-queries that target missing context. By traversing query-query edges and aggregating evidence across multiple hops, these methods significantly enhance the completeness of the retrieved context, particularly for questions requiring the synthesis of disjointed information scattered across diverse modalities.

The synergy between adaptive routing and iter-

ative refinement enables robust performance even under constraints of model size or index latency. While naive approaches suffer from performance degradation as hop counts increase, hybrid strategies that preserve hierarchical context and capture cross-section dependencies demonstrate superior resilience in handling deep reasoning chains. Furthermore, balancing effectiveness with efficiency remains critical; techniques such as limiting refinement rounds or employing approximate nearest neighbor search allow systems to maintain low latency without sacrificing the accuracy gains provided by deep, iterative exploration. Ultimately, this co-design of agent planning and dynamic retrieval fosters a more organic interaction between the query formulation and the knowledge base, moving beyond rigid pipelines toward flexible, context-aware information seeking.

2.3.2 Multi-Hop Reasoning via Knowledge Graph Integration

Multi-hop reasoning via knowledge graph integration addresses the limitations of unimodal retrieval by structuring scattered information into entity-centric networks that facilitate complex traversal. Traditional graph-based approaches typically rely on text-only nodes, extracting entity relations to support hierarchical exploration and logical chaining. However, these methods often struggle with long-range dependencies across disjointed content, as they lack the structural grounding necessary to connect semantically related but physically distant knowledge fragments. By leveraging knowledge graphs, systems can move beyond simple similarity search to perform explicit multi-hop traversal, effectively bridging gaps between isolated data points through predefined or dynamically inferred relational paths.

Recent advancements extend this paradigm to multimodal contexts, integrating visual cues such as charts, tables, and diagrams directly into the graph topology. Instead of treating visual elements as separate entities, modern frameworks encode them as structured nodes linked via semantic edges to textual descriptions and underlying data values. This integration allows for discrete reasoning over visually rich documents, where the graph captures both local chunk associations and global semantic neighborhoods. Techniques like semantic-oriented hierarchical graphs enable the model to preserve header contexts in multi-page tables and trace process flows in diagrams, ensur-

ing that reasoning chains remain consistent across different modalities during retrieval and synthesis.

To further enhance inference accuracy, contemporary systems employ consistency constraints and iterative refinement mechanisms within the graph structure. By generating intermediate reasoning steps, such as decomposed sub-questions anchored to specific graph nodes, models can validate evidence curation before final answer generation. This approach mitigates the degradation of performance often observed in high-hop queries by maintaining hierarchical context through sibling packing and capturing cross-section dependencies. Consequently, knowledge graph integration not only supports complex, cross-modal synthesis but also provides a transparent framework for verifying logical coherence, thereby improving robustness in domain-specific and agentic deep research workflows.

2.3.3 Noise Reduction through Adaptive Page Selection

Noise reduction in long-document processing is critical, as static top- k retrieval often saturates the context window with irrelevant pages, diluting salient evidence. Traditional methods relying on query-agnostic document representations fail to adapt to the uncertain distribution of relevant information across varying document lengths. To mitigate this, adaptive page selection mechanisms dynamically adjust the retrieval scope, filtering out visual noise such as blank pages, logos, and repetitive headers before evidence integration. By employing clustering algorithms like DBSCAN on layout elements, systems can identify and remove consistent structural artifacts that interfere with semantic matching, ensuring that only content-rich segments proceed to the reasoning stage.

Advanced frameworks implement query-adaptive strategies that go beyond simple similarity scoring to evaluate the informational density of candidate pages. These approaches utilize layout-aware parsing to detect explicit section headings and implicit topic shifts, enabling a "zoom-in" strategy where agents locate specific visual elements like tables or charts within retrieved pages. Rather than fixed window sizes, adaptive quantity mechanisms modulate the number of selected pages based on the semantic richness of the text and image modalities. This dynamic adjustment prevents the inclusion of low-value distractors while maintaining sufficient

contextual continuity for cross-page reasoning, effectively balancing coverage and precision without overwhelming the generator’s capacity.

Empirical analyses demonstrate that such adaptive selection significantly outperforms fixed baselines, particularly in scenarios involving high page counts where performance typically degrades due to noise accumulation. By restricting the input to evidence-dense multimodal contexts, these methods achieve higher accuracy with fewer context segments, reducing the irrelevant pages ratio while maintaining perfect recall. The synergy between retrieval filtering, decomposition, and layout-aware routing allows models to handle complex, multi-source documents more robustly. Consequently, adaptive page selection serves as a foundational component for optimizing lightweight multimodal large language models, enhancing both inference efficiency and answer fidelity in visually rich document understanding tasks [6].

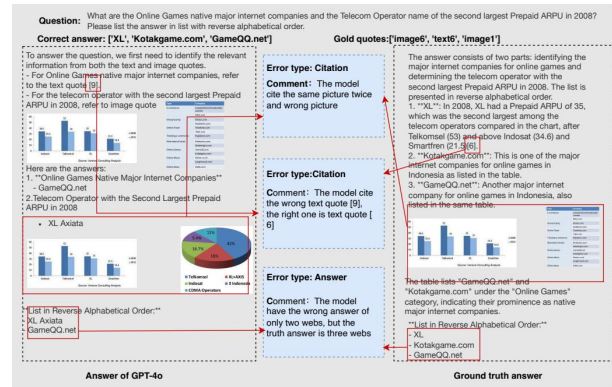


Figure 6: Chart from 'Benchmarking Retrieval-Augmented Multimodal Generation for Document Question Answering' (arXiv:2505.16470)

3 Agentic Frameworks and Reasoning Architectures

3.1 Collaborative Multi-Agent Orchestration

3.1.1 Role-Based Specialization and Collective Decision Making

Role-based specialization in multimodal Retrieval-Augmented Generation (RAG) systems addresses the heterogeneity of industrial documents by assigning distinct functional responsibilities to specific model agents. Rather than relying on a monolithic architecture, this paradigm decomposes the pipeline into specialized roles tailored for visually rich documents, such as layout analysis, se-

mantic indexing, and cross-modal reasoning. This taxonomy ensures that each component optimizes for specific retrieval granularities and information fidelity, effectively mitigating the limitations of single-model approaches that often struggle with the diverse structural complexities found in domain-specific knowledge bases.

Collective decision-making mechanisms orchestrate these specialized agents through dynamic routing and consensus protocols to resolve complex queries. Advanced frameworks employ question-level routers that analyze input characteristics to dispatch tasks to the most competent agent variants, enabling adaptive workflows that switch between global context synthesis and fine-grained element extraction. To integrate disparate outputs, systems utilize fusion strategies like Reciprocal Rank Fusion or attention-based combination methods, which balance conflicting signals from different retrieval paths. This coordinated approach allows the system to leverage the unique strengths of each role, ensuring robust performance across varying document types and query difficulties.

However, implementing such distributed architectures introduces a critical trade-off between system complexity and operational efficiency. While agentic workflows and multi-round self-reflection enhance reasoning capabilities, they often incur significant latency and computational costs that may yield only marginal performance gains in production environments. Consequently, effective deployment requires careful calibration of the number of specialized roles and the sophistication of the decision-making logic. Future research must focus on optimizing these collective mechanisms to maintain high accuracy without compromising the responsiveness required for real-time industrial applications, ensuring that the benefits of specialization do not come at the expense of scalability.

3.1.2 Iterative Feedback Loops and Self-Correction Mechanisms

Iterative feedback loops in retrieval-augmented generation systems function by dynamically refining query representations through a cyclic interaction between search and verification modules [7]. Rather than relying on static initial retrieval, these mechanisms decompose complex queries into sub-queries, executing multiple rounds of evidence gathering where each iteration incorporates feedback from previous retrieval failures. This process

often involves a dedicated Refiner agent that analyzes the sufficiency and relevance of retrieved context, generating modified search terms to address information gaps or resolve ambiguities identified in prior steps.

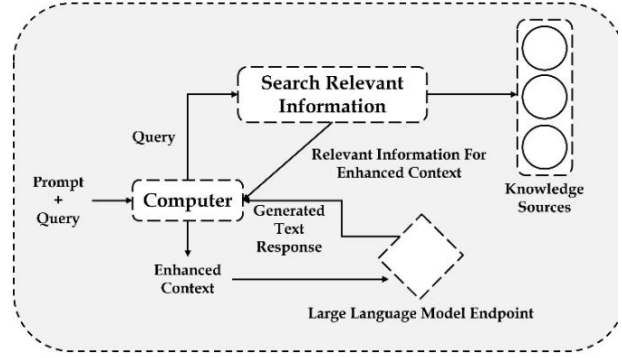


Figure 7: Chart from 'Conversational Text Extraction with Large Language Models Using Retrieval-Augmented Systems' (arXiv:2501.09801)

Self-correction mechanisms complement this iterative search by introducing internal verification stages that validate generated responses against source evidence before final output. These systems employ probabilistic frameworks or rule-based validators to detect hallucinations, logical inconsistencies, or citation errors, triggering automatic regeneration when confidence scores fall below predefined thresholds. By aggregating results from diverse retrieval frameworks or utilizing self-consistency sampling, models can mitigate individual retrieval failures and reduce error propagation across cascaded processing stages, effectively compensating for the limitations of single-pass generation pipelines.

Despite their efficacy in enhancing accuracy, these iterative approaches introduce significant computational overhead due to repeated large language model invocations and increased token consumption. The trade-off between performance gains and latency becomes critical, as diminishing returns are often observed beyond two or three refinement cycles. Consequently, recent advancements focus on optimizing these loops by freezing generator parameters during specific training phases or employing lightweight verifier models, aiming to balance the robustness of multi-step reasoning with the efficiency required for real-time deployment in resource-constrained environments.

3.1.3 Modular Tool Integration and Action Planning

Modular tool integration in multimodal RAG systems decomposes complex reasoning into distinct, specialized agents responsible for planning, evidence extraction, and answer synthesis. By separating these functions into chain-of-thought workflows, frameworks can address multi-hop queries and ambiguous contexts without requiring extensive fine-tuning. This architectural separation allows individual components to leverage domain-specific capabilities, such as layout-aware segmentation or cross-modal alignment, ensuring that retrieval and generation processes are optimized independently before being unified through a coherent action plan.

Advanced implementations treat these modular components as cooperative agents within a shared reward structure, utilizing reinforcement learning to align local decisions with global task objectives. Dynamic routing mechanisms further enhance this paradigm by automatically identifying task characteristics to activate appropriate model pathways, such as switching between high-reasoning modes for complex synthesis or lightweight retrievers for simple fact-checking. This flexibility enables the system to adaptively select tools based on query complexity, effectively balancing computational overhead against the need for rigorous verification and reflection across diverse document structures.

Despite the improved robustness and factuality offered by agent-based coordination, these systems introduce significant latency and economic costs due to repeated large language model invocations for planning and validation steps. The trade-off between the granularity of modular decomposition and inference efficiency remains a critical challenge, particularly in industrial deployments requiring real-time responses. Future research must focus on optimizing inter-agent communication protocols and developing lightweight planning strategies that maintain high reasoning fidelity while minimizing the computational burden inherent in multi-pass, tool-augmented pipelines.

3.2 Advanced Reasoning Paradigms and Cognitive Simulation

3.2.1 Chain-of-Thought Decomposition and Hypothesis Generation

Chain-of-Thought (CoT) decomposition serves as a critical mechanism for transforming complex,

multi-modal queries into manageable sub-tasks, thereby mitigating the cognitive load on large language models. By explicitly instructing models to outline a preliminary plan or break down questions into sequential reasoning steps, systems can effectively navigate intricate data structures such as nested tables and multi-dimensional plots. This structured approach prevents premature commitment to single interpretations, a common failure mode known as single-interpretation bias, and ensures that each logical step is grounded in specific retrieved evidence before proceeding to final synthesis.

Hypothesis generation operates synergistically with decomposition by encouraging the model to formulate multiple potential answers or semantic interpretations prior to evidence aggregation. Instead of merging conflicting signals from diverse sources, which often leads to interpretation conflation and hallucinated composite claims, the model generates distinct hypotheses tied to specific document chunks or visual elements. This process facilitates a more rigorous verification stage where candidates are evaluated against the source material, allowing the system to discard unsupported assertions and prioritize those with strong semantic alignment and factual grounding.

The integration of these strategies within Retrieval-Augmented Generation frameworks significantly enhances the faithfulness and accuracy of responses in fact-intensive domains [7]. By separating the reasoning trajectory from the final answer generation, models maintain a coherent working memory across extended contexts, effectively tracking multiple pieces of evidence without losing track of the original query intent. Furthermore, embedding self-verification steps within the CoT structure enables dynamic error correction, ensuring that the final output is not only logically sound but also strictly attributable to the provided context, thus reducing the propagation of fabricated details.

3.2.2 Probabilistic Evidence Corroboration and Belief Updating

Probabilistic evidence corroboration addresses the inherent uncertainty in multimodal retrieval by treating candidate evidence as stochastic variables rather than deterministic facts. Instead of relying on single-source retrieval, advanced frameworks aggregate signals from disjointed text chunks, tables, and visual regions to construct a unified be-

lief state. This approach mitigates hallucination by requiring consistent support across multiple modalities before elevating the confidence score of a potential answer. By modeling the joint probability distribution of retrieved items, systems can effectively bridge sparse information gaps, connecting distant data points that individually appear insufficient but collectively form a robust evidentiary path.

Belief updating mechanisms dynamically refine these probabilities as new evidence is processed through iterative reasoning loops. When a Vision-Language Model verifies a candidate pair, it performs a semantic alignment check that acts as a Bayesian update, adjusting the posterior probability of the hypothesis based on the likelihood of entailment versus contradiction. This process is critical for handling conflicting information, where the system must downgrade beliefs associated with misaligned visual grounding or inconsistent textual proxies. Fine-grained verification modules ensure that only candidates with strong cross-modal coherence contribute to the final decision, preventing the propagation of errors from noisy retrievers into the generation phase.

The integration of uncertainty-aware fusion further enhances reliability by explicitly quantifying the confidence bounds of aggregated evidence. In high-stakes domains such as legal or financial analysis, this probabilistic framework allows the system to abstain from answering when the cumulative belief fails to meet a rigorous threshold, thereby prioritizing faithfulness over completeness. By maintaining a transparent trail of how individual evidence items influence the global belief state, these methods provide verifiable attribution for every generated claim. Consequently, the shift from deterministic retrieval to probabilistic corroboration represents a fundamental advancement in creating trustworthy document understanding systems capable of navigating complex, multi-page contexts.

3.2.3 Reflective Reasoning and Memory Consolidation

Reflective reasoning mechanisms have emerged as a critical component for enhancing the reliability of multimodal retrieval-augmented generation [8]. By integrating self-critique loops, models can dynamically evaluate the sufficiency of retrieved visual and textual evidence before committing to a final response. This iterative process al-

lows systems to identify gaps in context or potential hallucinations, triggering additional retrieval steps or re-weighting existing information to ensure factual consistency. Such self-reflection transforms static generation into an adaptive dialogue between the model’s internal knowledge and external data sources.

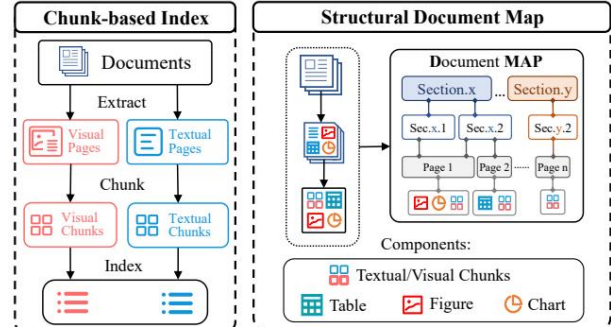


Figure 8: Chart from 'DMAP: Human-Aligned Structural Document Map for Multimodal Document Understanding' (arXiv:2601.18203)

To support these complex reasoning trajectories, advanced memory consolidation strategies are employed to maintain coherence across extended interaction sequences. Shared memory architectures and caching mechanisms store intermediate reasoning states and verified facts, reducing redundant computation and preventing context drift during multi-step problem solving. Lightweight proxy models often serve as early-stage filters within this framework, deciding when to invoke heavy-weight reasoning modules based on task uncertainty. This hierarchical approach optimizes computational resources while preserving the fidelity of long-term dependencies required for intricate visual question answering.

The synergy between reflective reasoning and memory consolidation significantly improves performance on tasks demanding deep cross-modal understanding. Empirical observations indicate that models leveraging these techniques exhibit reduced modality skew and higher grounding precision, particularly when faced with conflicting evidence or distractors. By explicitly tracking reasoning chains and consolidating verified insights into a persistent memory state, systems achieve greater stability and interpretability. Consequently, this paradigm shifts the focus from mere pattern matching to genuine cognitive processing, enabling more robust handling of visually rich information in dynamic environments.

3.3 Evaluation Protocols and Robustness Assurance

3.3.1 LLM-as-Judge Frameworks for Answer Verification

LLM-as-Judge frameworks have emerged as a critical mechanism for verifying answer fidelity in Retrieval-Augmented Generation, addressing the limitations of traditional string-matching metrics like ROUGE-L which often fail to capture semantic correctness. These systems leverage powerful language models to evaluate generated responses against ground truth or retrieved evidence on multi-dimensional scales, typically assessing faithfulness, completeness, and factual consistency. By employing dedicated prompts, judges assign scalar scores or binary labels that reflect whether the core information is accurate and strictly grounded in the provided context, thereby offering a more robust assessment of hallucination rates and logical coherence in complex reasoning tasks.

Recent advancements extend this paradigm into the multimodal domain, where Vision-Language Models serve as specialized verifiers for cross-modal evidence alignment. Unlike unimodal judges, these frameworks perform fine-grained verification to ensure that textual claims are explicitly supported by corresponding visual elements, such as charts, tables, or document layouts. This capability is essential for detecting subtle discrepancies where text may be semantically relevant but visually unsupported, effectively filtering false positives through a process of claim-level entailment. Such modules often operate post-retrieval to re-rank candidates based on semantic agreement, ensuring that only high-confidence, contextually faithful results contribute to the final output.

Despite their efficacy, LLM-as-Judge approaches face challenges regarding stability and bias, as evaluation scores can fluctuate based on prompt formatting, retrieval depth, and the specific backbone model employed [9]. There is a recognized divergence between automated judge assessments and expert human evaluations, particularly in domain-specific scenarios requiring nuanced understanding. Consequently, current research emphasizes the development of risk-aware trust calibration and standardized evaluation protocols that mitigate single-model bias, often utilizing ensemble strategies or multi-model consensus to enhance the reliability of automated verification

in production environments.

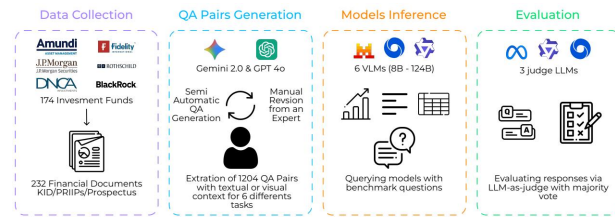


Figure 9: Chart from 'When Tables Go Crazy: Evaluating Multimodal Models on French Financial Documents' (arXiv:2602.10384)

3.3.2 Hallucination Mitigation via Source Attribution

Hallucination mitigation via source attribution fundamentally shifts the generation paradigm from opaque synthesis to transparent, evidence-grounded reasoning. By enforcing explicit links between generated claims and specific segments of retrieved context, systems can significantly reduce the propagation of misinformation inherent in large vision-language models. This approach operates on the premise that restricting the input context to fine-grained, relevant blocks rather than entire pages allows the model to operate on structurally grounded evidence, thereby minimizing the inclusion of irrelevant information that often triggers factual deviations.

Advanced implementations extend this concept beyond simple text citation to include precise visual grounding within multimodal documents. Techniques such as bounding box generation or direct image cropping enable models to highlight exact regions in charts, tables, or diagrams that support a specific statement. This granular attribution not only enhances the verifiability of the output but also forces the model to align its internal reasoning with observable data features. Consequently, methods that integrate spatial coordinates and layout information into the retrieval pipeline demonstrate superior performance in filtering out high-similarity visual hallucinations compared to traditional document-level retrieval strategies.

Despite these advancements, achieving robust statement-level grounding remains a complex challenge, particularly when dealing with heterogeneous data sources. Current frameworks often struggle with block-level citation accuracy, frequently defaulting to section-level attribution which lacks the precision required for rigorous fact-checking. Future directions emphasize the de-

velopment of verification mechanisms that jointly evaluate cross-modal agreement between textual and visual evidence paths. By quantifying the consistency between different modalities and penalizing conflicting evidence, next-generation systems aim to establish a more reliable foundation for automated knowledge synthesis in visually rich environments.

3.3.3 Benchmarking Complex Multi-Document Reasoning

Early benchmarks primarily focused on single-page visual question answering, limiting the assessment of models to localized extraction tasks without requiring cross-page synthesis. To address this, recent datasets have expanded scope to multi-page and multi-document settings, introducing challenges that demand hierarchical parsing and long-context retention. These evolved benchmarks necessitate that systems not only identify relevant pages within extensive corpora but also integrate disparate visual and textual elements scattered across different sections to construct coherent answers.

Current evaluation frameworks increasingly emphasize multi-hop reasoning capabilities, where answers cannot be derived from a single evidence snippet but require chaining information from tables, charts, and narrative text across multiple documents [10]. Specialized datasets now incorporate consistency-constrained modality fusion to test unified reasoning steps, ensuring models can resolve conflicts between visual data and textual descriptions. Furthermore, these benchmarks introduce adaptive retrieval strategies to measure how effectively systems recover fine-grained evidence from visually rich, heterogeneous document collections under realistic noise conditions.

Despite these advancements, significant gaps remain in evaluating meta-level reasoning and complex numerical inference across diverse domains such as finance and engineering. Existing metrics often struggle to quantify the precision of cross-document attribution or the robustness of models against layout variations and low-quality OCR. Future benchmarking efforts must therefore prioritize scalable, multilingual evaluations that rigorously test higher-order information synthesis, moving beyond simple retrieval accuracy to assess the depth of logical deduction and structural understanding in complex multi-document environments.

4 Chart Understanding and Visualization Generation

4.1 Semantic Perception and Data Extraction

4.1.1 OCR-Free Visual Encoding and Element Detection

Recent advancements in multimodal understanding have catalyzed a paradigm shift toward OCR-free visual encoding, aiming to bypass the error propagation inherent in traditional optical character recognition pipelines. Instead of relying on explicit text extraction, modern architectures leverage high-resolution Vision Transformers (ViT) capable of directly processing raw pixel data to capture fine-grained textual and graphical features simultaneously. Techniques such as parameter-free cropping modules partition high-resolution inputs into manageable sub-images, enabling encoders to resolve small text instances without dedicated OCR engines. This approach facilitates a unified representation where visual semantics and textual content are embedded within a shared latent space, allowing models to reason over chart elements holistically rather than through disjointed text-image fusion.

Parallel to encoding improvements, element detection methodologies have evolved from coordinate regression to sophisticated grounding mechanisms that map natural language queries directly to visual regions. Contemporary frameworks employ set-of-marks prompting and detection transformers trained on diverse chart corpora to localize data marks, axes, and legends with high precision. By treating detection as a multimodal grounding task, these systems identify structural components like bars or pie slices based on semantic intent rather than rigid geometric heuristics. This capability is crucial for attributing specific data values to visual glyphs, ensuring that downstream reasoning is anchored in accurate spatial contexts without intermediate textual conversion steps.

The integration of OCR-free encoding with precise element detection significantly enhances robustness in zero-shot chart analysis and generation tasks. By eliminating the dependency on external OCR tools, models mitigate failures caused by low-resolution inputs, complex layouts, or rotated text, which traditionally degrade performance [11]. Furthermore, this end-to-end visual processing supports scalable retrieval and reasoning over large document collections, where maintaining fidelity between visual evidence and gener-

ated insights is paramount. Consequently, the field is moving towards unified architectures that treat charts as native visual-language entities, enabling more reliable attribution and deeper semantic understanding of complex data visualizations.

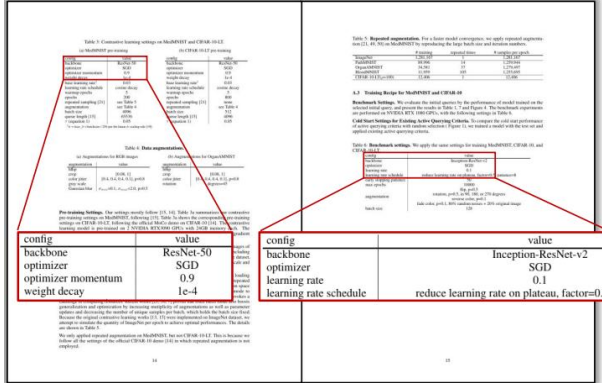


Figure 10: Chart from 'Docopilot: Improving Multimodal Models for Document-Level Understanding' (arXiv:2507.14675)

4.1.2 Structural Graph Modeling for Chart Components

Structural graph modeling has emerged as a pivotal technique for decomposing charts into semantically rich components, addressing the limitations of purely visual or textual analysis. By defining explicit entities such as axes, legends, data marks, and labels, alongside relationships denoting hierarchy, relative position, and logical grouping, these models construct a comprehensive scene graph representation. This structured approach facilitates the precise extraction of underlying data tables and visual channels, enabling systems to distinguish between structural information and content-specific details with greater accuracy than unstructured methods.

Recent advancements leverage these graph structures to enhance multimodal reasoning and question-answering capabilities [8]. Instead of relying on flat feature vectors, relation-based graphs capture the intricate dependencies between chart elements, such as the connection between a specific data series and its corresponding legend entry or the alignment of ticks with axis values. Empirical evidence suggests that incorporating such relational inductive biases significantly outperforms fully connected graph alternatives, as the former preserves the semantic topology essential for interpreting complex visualizations. This structural fidelity is particularly critical when handling diverse

chart types, where the spatial arrangement directly encodes quantitative meaning.

Furthermore, structural graphs serve as a robust intermediate representation for bridging the gap between raw pixel data and high-level natural language descriptions. By parsing charts into standardized graph formats, models can effectively align visual encodings with textual contexts, supporting tasks ranging from automated captioning to fact-checking via citation localization. This modular representation not only improves the interpretability of chart understanding systems but also enhances their generalizability across different domains and layouts [12]. Consequently, structural graph modeling provides a scalable foundation for developing next-generation chart analysis tools capable of sophisticated logical reasoning and context-aware interaction.

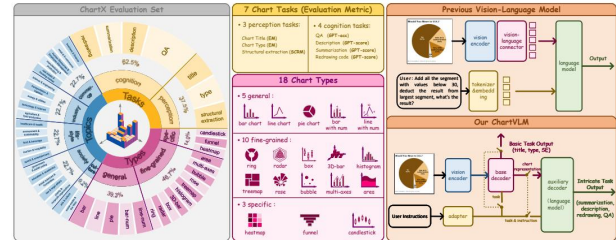


Figure 11: Chart from 'ChartX & ChartVLM: A Versatile Benchmark and Foundation Model for Complicated Chart Reasoning' (arXiv:2402.12185)

4.1.3 Table Distillation and Numerical Value Recovery

Table distillation serves as a critical intermediate step in chart understanding, aiming to reconstruct the underlying structured data from visual representations. This process typically involves converting graphical elements into tabular formats before generating descriptive text or answering queries. While effective for capturing explicit data points, standard distillation approaches often suffer from significant information loss regarding chart types, curve trends, and semantic relationships. Consequently, recent methodologies have shifted towards preserving these structural nuances during the conversion process to ensure that the distilled tables remain faithful to the original visual insights.

Numerical value recovery addresses the specific challenge of accurately extracting and verifying quantitative data within these reconstructed tables. Techniques frequently leverage Named

Entity Recognition modules to identify numerical phrases, such as percentages, monetary values, and cardinals, before cross-referencing them against the chart’s axis scales [13]. However, reliance on heuristic templates for identifying these values often introduces false positives, particularly when numerical phrases in accompanying text do not directly correspond to the visualized data. Robust recovery mechanisms must therefore integrate visual channel analysis to validate extracted numbers against the coordinate systems of the source graphics.

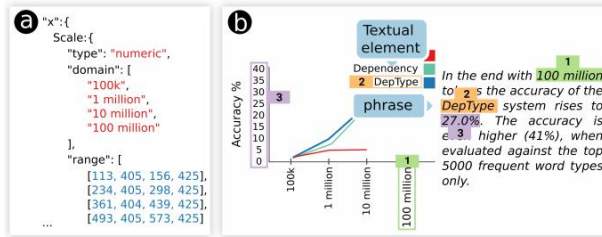


Figure 12: Chart from 'ChartText: Linking Text with Charts in Documents' (arXiv:2201.05043)

To mitigate errors in numerical extraction, advanced pipelines incorporate refinement stages that compare extracted values with domain constraints and utilize chain-of-thought reasoning for intermediate calculations. Ablation studies indicate that removing chart refinement processes or exemplar learning leads to substantial performance degradation, highlighting the necessity of iterative verification. Furthermore, integrating semantic insights during distillation allows models to better handle complex queries by distinguishing between accurate data regions and potential artifacts, ultimately enhancing the reliability of downstream tasks like question answering and data synthesis.

4.2 Analytical Reasoning and Insight Synthesis

4.2.1 Trend Identification and Comparative Analysis

Trend identification in chart understanding has evolved from simple keyword matching to sophisticated multimodal reasoning capable of discerning complex temporal patterns. Contemporary approaches categorize trends into distinct classes such as linear continuities, cyclical fluctuations, and abrupt mutations, often leveraging normalized data representations to handle varying scales ranging from probabilities to astronomical fig-

ures. Advanced models now generate both natural language descriptions and numerical trend summaries, explicitly capturing directional shifts like sharp drops followed by gradual recoveries, thereby bridging the gap between raw pixel data and semantic interpretation.

Comparative analysis extends beyond single-chart interpretation to involve cross-chart reasoning, where systems must identify diverging or converging trajectories across multiple visualizations. This process frequently requires resolving abstract queries that demand an understanding of relative changes rather than absolute values, a task where many current multimodal large language models exhibit performance degradation. Effective comparative frameworks employ specialized metrics, such as curve trend consistency scores and Fréchet distance, to quantify the alignment between predicted and ground-truth trajectories, ensuring that subtle deviations in slope or volatility are accurately detected and reported.

Despite these advancements, significant challenges remain in handling abstract data concepts and maintaining accuracy across diverse chart types like line, bar, and area graphs. Errors often arise when models fail to generalize trend semantics or misinterpret the magnitude of change due to a lack of unit-aware evaluation metrics. Future research directions emphasize the integration of domain-specific knowledge and refined fine-tuning strategies to improve the robustness of trend detection, particularly in zero-shot settings where models must infer complex relationships without explicit training examples.

4.2.2 Multi-Chart Reasoning and Cross-Visual Integration

Multi-chart reasoning necessitates moving beyond isolated visual analysis to establish semantic correlations across diverse diagrams within a single document [14]. Traditional approaches often fail to preserve individual chart distinctions, leading to hallucinated connections or lost context when synthesizing information from multiple sources. Recent advancements address this by constructing high-quality datasets that explicitly link figures, tables, and textual captions, enabling models to learn joint comprehension strategies. These frameworks leverage structural cues, such as document outlines or scene graphs, to maintain consistency and guide the model in identifying relevant data points across disparate visual elements.

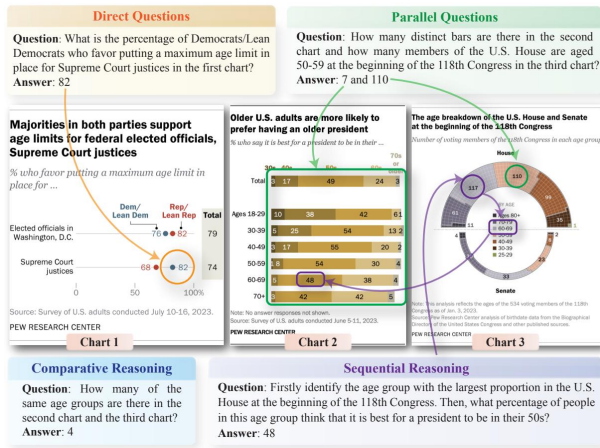


Figure 13: Chart from 'MultiChartQA: Benchmarking Vision-Language Models on Multi-Chart Problems' (arXiv:2410.14179)

Cross-visual integration further complicates this landscape by requiring the alignment of heterogeneous visual encodings, ranging from statistical plots to schematic diagrams. Effective systems now employ multi-head encoders and specialized attention mechanisms to simultaneously process normalized data scales and geometric parameters without conflating distinct chart types. By decomposing user intent into iterative extraction and refinement stages, these architectures can bridge the gap between low-level pixel features and high-level semantic meanings. This allows for the generation of executable code or natural language summaries that accurately reflect complex, multi-step logical dependencies found in scientific literature.

Despite these strides, significant challenges remain in handling open-ended scenarios where visual evidence is sparse or contradictory. Current benchmarks often rely on synthetic data or limited question types, failing to capture the nuanced reasoning required for real-world document analysis. Future research must focus on developing robust evaluation metrics that assess not only data extraction accuracy but also the correctness of cross-chart logical inference. Enhancing the interoperability between visual perception modules and large language models will be crucial for creating conversational agents capable of navigating intricate, multi-modal scientific narratives with human-level proficiency [14].

4.2.3 Fact-Checking and Hallucination Detection in Visual Data

Fact-checking in visual data necessitates bridging the semantic gap between generated textual descriptions and the underlying quantitative evidence within charts. Unlike standard natural language verification, this domain requires models to decode complex mappings between visual elements—such as axes, legends, and data points—and their corresponding numerical values. Current methodologies often struggle with this alignment, leading to plausible yet factually incorrect assertions where the text contradicts the visualized trends. Effective detection systems must therefore move beyond surface-level textual consistency to perform deep structural parsing, ensuring that every claim is strictly grounded in the source-supported values depicted in the visualization.

To mitigate hallucinations, recent advances leverage contextual embeddings and specialized visual encoders to enhance grounding capabilities. Techniques involving scene graphs jointly capture structural and semantical knowledge, allowing models to reason over the relationships between chart components rather than relying solely on pixel-level features. By integrating cross-attention mechanisms, these systems condense visual features into robust representations that can be cross-referenced against generated captions. This approach significantly reduces the incidence of "unconvincing visuals" and artificial data patterns, as the model is constrained to interpret only those features explicitly present in the input image, thereby minimizing speculative content generation.

Despite these improvements, significant challenges remain in handling deceptive visualizations and ensuring robust uncertainty estimation [15]. Existing metrics often fail to penalize hallucinations adequately when the generated text shares high lexical similarity with references but lacks visual relevance. Future directions emphasize the development of user-in-the-loop validation frameworks and multi-head encoders capable of distinguishing between authentic and perturbed data distributions. Furthermore, addressing the inability of current large multimodal models to perform complex visual-based reasoning over external references is critical [16]. Establishing rigorous safeguards that combine automated detection with human oversight is essential for the responsible de-

ployment of automatic charting and analysis systems in high-stakes environments.

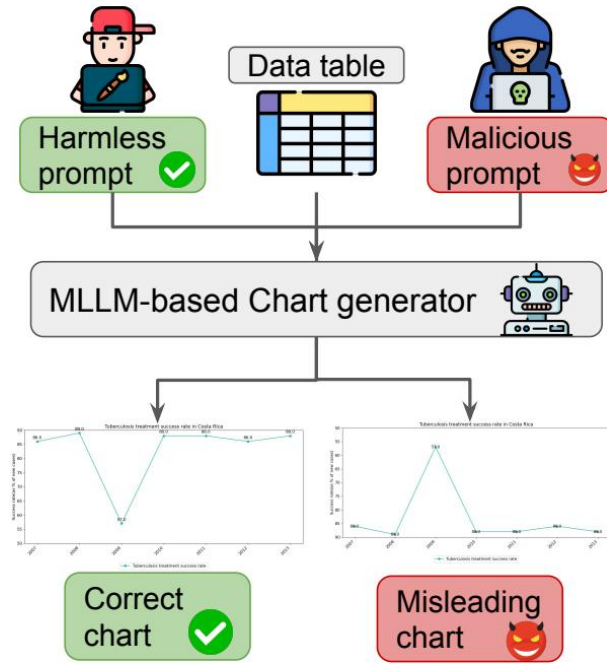


Figure 14: Chart from 'ChartAttack: Testing the Vulnerability of LLMs to Malicious Prompting in Chart Generation' (arXiv:2601.12983)

$$\mathcal{L}_{\text{MRL}}(q, d^+) = \sum_{m_i \in M} w_i \cdot \ell \left(\mathcal{E}(q)^{[m_i]}, \mathcal{E}(d^+)^{[m_i]} \right)$$

Figure 15: Chart from 'Unlocking Multimodal Document Intelligence: From Current Triumphs to Future Frontiers of Visual Document Retrieval' (arXiv:2602.19961)

4.3 Generative Visualization and Narrative Construction

4.3.1 Code-Driven Chart Synthesis and Style Transfer

Code-driven chart synthesis has emerged as a dominant paradigm for transforming abstract natural language into precise, executable visualizations. Modern frameworks leverage large language models to decompose user intent through unsupervised, multi-stage processes that iteratively extract relevant data and select optimal chart types based on visualization heuristics. By generating code in libraries such as Matplotlib in a zero-shot manner, these systems bypass manual coding requirements while ensuring high data accuracy. This approach uniquely connects document text, captions, and underlying data structures to maintain contextual

consistency, effectively addressing challenges related to varying data scales and heterogeneous input shapes through specialized pre-processing and normalization steps.

Beyond basic generation, recent advancements focus on expressive style transfer and parameter randomization to enhance visual diversity and aesthetic appeal. Techniques involve systematically varying visualization parameters, such as alternating line styles between solid and dashed or modifying marker shapes in scatter plots, to create rich datasets for training robust models. More sophisticated methods enable structural transformations, such as converting line charts into area charts or pie charts into donut variants, by adjusting single parameters within the generated code. These capabilities allow for the automated augmentation of standard charts with intuitive visual encodings, ensuring that the final output not only reflects the data accurately but also adheres to specific stylistic constraints or design preferences.

The integration of code generation with style transfer facilitates complex applications ranging from scientific diagram analysis to interactive infographic design. By orchestrating specialized agents to extract structured tables, break down logical steps, and generate contextual descriptions, these systems achieve superior performance in visual understanding tasks compared to direct prompting baselines. Furthermore, the ability to reverse-engineer visual encodings from bitmap images enables the repurposing and redesign of existing visualizations without access to original source files. This synergy between generative code and flexible styling mechanisms establishes a robust foundation for creating adaptive, high-fidelity visualizations that align closely with user intentions across diverse domains.

4.3.2 Data-Driven Storytelling and Text-Chart Interleaving

Data-driven storytelling has evolved from simple captioning to complex narrative construction where text and charts are dynamically interleaved to convey analytical insights. Unlike traditional text-to-chart tasks that rely on explicit, structured inputs, modern approaches must extract latent data from lengthy, unstructured documents to generate coherent visual narratives. This shift necessitates models capable of distinguishing relevant statistical signals from noise within dense textual contexts, moving beyond direct mapping to perform

sophisticated data inference and semantic alignment between narrative flow and visual representation.

Current methodologies increasingly leverage Multimodal Large Language Models to bridge the gap between documentary text and visual encoding [17]. These systems employ specialized extraction modules to convert unstructured descriptions into intermediate tabular representations, which then serve as precise prompts for chart generation engines. By integrating chain-of-thought reasoning, advanced frameworks can decompose complex analytical intents into logical steps, ensuring that generated charts not only reflect accurate numerical values but also maintain contextual consistency with the surrounding explanatory text, thereby supporting multi-hop reasoning across document sections.

$$M^* = \left\{ (k^*, l) \mid k^* = \underset{(k,l) \in \mathcal{E}}{\operatorname{argmax}} w_{k,l}, \forall l \in [n] \right\}.$$

Figure 16: Chart from 'MRAMG-Bench: A Comprehensive Benchmark for Advancing Multimodal Retrieval-Augmented Multimodal Generation' (arXiv:2502.04176)

Despite these advancements, significant challenges remain in achieving robust visual-language matching and handling multi-chart scenarios within single documents. Existing benchmarks often focus on isolated chart interactions, failing to capture the complexity of real-world scientific papers where figures, tables, and text must be synthesized holistically. Future research directions emphasize the development of unified architectures that can manage long-context dependencies, resolve ambiguities in element localization, and provide explainable connections between specific textual claims and their corresponding visual evidence, ultimately enhancing user trust and analytical productivity.

4.3.3 Automated Metadata Generation and Accessibility Enhancement

Automated metadata generation has emerged as a critical mechanism for bridging the gap between raw document structures and semantic understanding. Contemporary pipelines leverage large language models to synthesize rich metadata from unstructured inputs, moving beyond simple label extraction to capture complex spatial relationships

and visual attributes. This process typically involves decomposing tabular or graphical data into distinct facets, which are then mapped to fixed-size visualization references. By automating the annotation of text roles and layout comprehension, these systems facilitate location-aware information extraction and enable sophisticated downstream tasks without relying on labor-intensive manual curation.

The integration of automated metadata directly enhances accessibility by transforming static visualizations into semantically rich, machine-readable descriptions. Traditional template-based approaches often fail to convey the nuanced analytical insights present in scientific charts, whereas modern methods employ multi-level semantic modeling to generate comprehensive natural language alternatives. These systems analyze data patterns to produce descriptive tags covering definitions, usage scenarios, and data characteristics, ensuring that visually impaired users receive contextually relevant information. Furthermore, this automation supports the creation of interactive overlays that link textual explanations with specific graphic elements, significantly reducing the effort required to author accessible content.

Despite these advancements, challenges remain in ensuring the robustness and fidelity of generated metadata across diverse document formats. Current research highlights the necessity of evaluating model performance against noise and varying layout complexities to prevent the propagation of misleading visualizations. Future directions emphasize the development of code-driven methods to avoid distributional convergence in generated content, ensuring that metadata accurately reflects the underlying data logic rather than superficial patterns. By refining these automated processes, the field aims to establish trustworthy multimodal systems that balance data utility with privacy while universally improving document accessibility.

5 Future Directions

Despite significant advancements in multimodal Retrieval-Augmented Generation for document and chart understanding, several critical limitations persist within current frameworks. Existing systems often struggle to maintain consistent spatial coherence when processing extremely long-context documents where visual and textual dependencies span dozens of pages, leading to frag-

mented reasoning chains. Furthermore, while late interaction and generative embedding strategies have reduced reliance on brittle Optical Character Recognition pipelines, they remain computationally expensive, hindering real-time deployment in resource-constrained environments. A notable gap also exists in the robust handling of dynamic or interactive visualizations, as most current models are trained exclusively on static image representations, failing to capture the temporal evolution of data or the underlying logic of animated charts. Additionally, the lack of standardized benchmarks specifically designed for fine-grained, layout-rich multi-hop reasoning makes it difficult to objectively compare the efficacy of different structural awareness techniques across diverse document domains.

Future research should prioritize the development of efficient, hierarchical attention mechanisms that can dynamically prune irrelevant visual patches without sacrificing the precision required for fine-grained alignment. Investigating neuro-symbolic approaches that combine the semantic flexibility of large language models with the rigorous logical constraints of symbolic graph traversal could significantly enhance multi-hop reasoning capabilities, ensuring that derived conclusions strictly adhere to the structural topology of the source document. Another promising direction involves the integration of temporal modeling to process dynamic visual content, enabling systems to interpret trends and changes in time-series visualizations rather than treating them as isolated static frames. Moreover, the community must focus on creating self-supervised learning objectives that leverage vast amounts of unlabeled document data to improve cross-modal alignment, reducing the dependency on costly, manually annotated datasets for training layout-aware retrievers. Finally, exploring adaptive retrieval policies that automatically switch between lightweight lexical search and heavy graph-based traversal based on real-time query complexity analysis will be essential for optimizing computational resources.

The successful realization of these research directions promises to fundamentally transform document intelligence systems from passive information retrievers into active, reasoning agents capable of deep comprehension. By overcoming current bottlenecks in spatial coherence and computational efficiency, future frameworks will enable highly accurate, visually grounded responses for

complex professional domains such as legal contract analysis, scientific literature review, and financial auditing. This evolution will not only minimize hallucinations by ensuring every generated claim is traceable to specific visual and textual evidence but also democratize access to knowledge embedded in complex, non-linear documents. Ultimately, bridging these gaps will foster a new generation of AI assistants that can navigate the full spectrum of human knowledge representation, turning dense, multimodal archives into actionable, reliable insights.

6 Conclusion

This survey has systematically examined the evolution of Multimodal Retrieval-Augmented Generation frameworks specifically designed for the complex challenges of chart and document understanding. We analyzed the paradigm shift from early fusion methods to sophisticated late interaction mechanisms that preserve fine-grained token and patch-level details, enabling precise cross-modal alignment without the information loss inherent in global pooling. The review highlighted the emergence of generative embedding paradigms and hybrid sparse-dense fusion strategies, which collectively bridge the gap between user intent and heterogeneous visual-textual content. Furthermore, we explored structural awareness techniques, including layout-aware graph construction and hierarchical tree parsing, which decompose documents into interconnected nodes to maintain logical flow and spatial topology. Finally, the discussion on dynamic retrieval and adaptive context aggregation demonstrated how systems can now optimize computational resources by routing queries based on complexity, thereby facilitating robust multi-hop reasoning across disjointed visual and textual entities.

The significance of this work lies in its consolidation of disparate research threads into a unified roadmap for next-generation document intelligence. By critically categorizing state-of-the-art techniques ranging from vision-guided chunking to recursive retrieval loops, this survey elucidates how structural and visual nuances can be effectively leveraged to overcome the limitations of traditional text-centric retrieval systems. It provides an essential reference for researchers and practitioners aiming to build systems that do not merely process images as holistic units but in-

stead dissect and index fine-grained visual elements within complex layouts. This comprehensive synthesis identifies current bottlenecks in handling long-context, layout-rich data and offers a clear perspective on integrating semantic density with lexical precision, ultimately serving as a foundational guide for developing more accurate, visually grounded, and scalable knowledge systems.

Looking forward, the field must prioritize the development of unified architectures that seamlessly integrate spatial reasoning with semantic understanding to fully realize the potential of multimodal RAG. Future research should focus on refining adaptive mechanisms that dynamically balance retrieval depth with computational efficiency, particularly for extremely large-scale document repositories. There is also a critical need for standardized benchmarks that specifically evaluate fine-grained visual localization and multi-hop reasoning capabilities across diverse document types. We call upon the research community to move beyond brittle optical character recognition pipelines and embrace direct visual-textual alignment strategies that capture subtle correlations in data visualizations. By addressing these challenges, the next wave of innovation will deliver robust document intelligence systems capable of transforming static, multimodal repositories into dynamic, actionable knowledge sources.

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